

1/1 Internal Kink Mode and Kadomtsev Reconnection Model in Tokamak Plasmas

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I. EQUILIBRIUM PROFILE

Let a , R_0 , B_0 , q_0 , q_a , p_0 , and ϵ_a be the minor and major radii, toroidal magnetic field, central safety factor, edge safety factor, central pressure, and inverse aspect ratio a/R_0 , respectively. We define the normalized radius $\tilde{r} = r/a$ with $\tilde{r} \in [0, 1]$ and $\mu = q_a/q_0$. The equilibrium profile before the sawtooth crash $q_i(\tilde{r})$ is given by

$$q_i(\tilde{r}) = q_a \frac{\tilde{r}^2}{1 - (1 - \tilde{r}^2)^\mu}, \quad (1)$$

with $q_i(\tilde{r} \rightarrow 0) = q_0$, $q_i(\tilde{r} = 1) = q_a$; the pressure profile before the crash $p_i(\tilde{r})$ is given by

$$p_i(\tilde{r}) = p_0(1 - \tilde{r}^2)^\mu, \quad (2)$$

and the poloidal magnetic field $B_\theta(\tilde{r})$ is given by

$$B_\theta(\tilde{r}) = \frac{\tilde{r}B_0\epsilon_a}{q_i(\tilde{r})}. \quad (3)$$

In a tokamak, the magnetic field is

$$\mathbf{B} = B_\theta(\tilde{r})\hat{\boldsymbol{\theta}} + B_\phi(\tilde{r}, \theta)\hat{\boldsymbol{\phi}} \quad (4)$$

where $B_\phi(\tilde{r}, \theta) \approx B_0$ for large aspect ratio tokamaks with $r/R_0 \ll 1$. For the $m/n = 1/1$ mode, the poloidal field satisfies $B_\theta/B_0 = a\tilde{r}/R_0$, and we define the effective field

$$\mathbf{B}^*(\tilde{r}) = \mathbf{B} - \frac{B_0 r}{R_0} \hat{\boldsymbol{\theta}} = \left(B_\theta(\tilde{r}) - q(\tilde{r})B_\theta(\tilde{r}) \right) \hat{\boldsymbol{\theta}} + B_0 \hat{\boldsymbol{\phi}} \quad (5)$$

where

$$q(\tilde{r}) = \frac{\tilde{r}B_0\epsilon_a}{B_\theta(\tilde{r})} \quad (6)$$

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from Eq. 3.

We first study the poloidal component of the effective magnetic field $B_\theta^*(\tilde{r}) = B_\theta(\tilde{r})[1 - q(\tilde{r})]$ by introducing the effective poloidal flux $\psi^*(\tilde{r})$ with the given relation

$$\frac{d\psi^*}{d\tilde{r}} = aB_\theta^*(\tilde{r}) = aB_\theta(\tilde{r})[1 - q(\tilde{r})], \quad (7)$$

where such flux satisfies $\mathbf{B}^* = \nabla\psi^* \times \hat{\phi}$. The effective poloidal flux is obtained by

$$\psi^*(\tilde{r}) = a \int_0^{\tilde{r}} \frac{d\psi^*}{d\tilde{r}} d\tilde{r} = a \int_0^{\tilde{r}} [B_\theta(\tilde{r}') - \tilde{r}' B_0 \epsilon_a] d\tilde{r}'. \quad (8)$$

II. SAWTOOTH OSCILLATION AND KADOMTSEV RECONNECTION

The sawtooth instability occurs as the central safety factor $q(0)$ falls below 1; in other words, the crash is a characteristic internal kink mode on the $m/n = 1/1$ resonant surface. This instability exhibits a series of periodic ‘sawtooth’ drops in the plasma current, temperature, and pressure inside the instability boundary radius, shifting the magnetic axis. We are concerned with the ideal 1/1 kink mode scenario with the initial conditions $q_0 < 1$ and $q_a > 1$.

We numerically evaluate the radial coordinates as the instability occurs according to the model developed above. Consider

$$\Delta\tilde{r} = \frac{1}{N_r - 1}, \quad (9)$$

$$\tilde{r}_j = j(\Delta\tilde{r}), \quad (10)$$

$$q_i(\tilde{r}_j) = q_a \frac{(\tilde{r}^2)_j}{1 - [1 - (\tilde{r}^2)_j]^\mu}, \quad (11)$$

$$B_\theta(\tilde{r}_j) = \frac{\tilde{r}_j B_0 \epsilon_a}{q_i(\tilde{r}_j)}, \quad (12)$$

and

$$\left[\frac{d\psi^*}{d\tilde{r}} \right]_j = aB_\theta(\tilde{r}_j)[1 - q_i(\tilde{r}_j)]. \quad (13)$$

The effective flux ψ_j^* is calculated by

$$\psi^*(\tilde{r}_j) = \psi^*(\tilde{r}_{j-1}) + \frac{\Delta\tilde{r}}{2} \left(\left[\frac{d\psi^*}{d\tilde{r}} \right]_j + \left[\frac{d\psi^*}{d\tilde{r}} \right]_{j-1} \right). \quad (14)$$

To find the normalized resonance radius $\tilde{r}_1$ where the instability occurs; i.e., $\tilde{r}$ such that $B_\theta^*(\tilde{r}) = 0$, we evaluate $\tilde{r}_1$ where

$$q(\tilde{r}) - 1 = 0. \quad (15)$$

If there exists such $\tilde{r}_1$, the index k can be evaluated where

$$\left[\frac{d\psi^*}{d\tilde{r}} \right]_k \geq 0, \quad \left[\frac{d\psi^*}{d\tilde{r}} \right]_{k+1} < 0. \quad (16)$$

Then, $\tilde{r}_1$ may be obtained through linear interpolation

$$\tilde{r}_1 = \tilde{r}_k - \frac{\left[\frac{d\psi^*}{d\tilde{r}} \right]_k}{\left[\frac{d\psi^*}{d\tilde{r}} \right]_{k+1} - \left[\frac{d\psi^*}{d\tilde{r}} \right]_k} \Delta \tilde{r}_k. \quad (17)$$

The internal region is bounded by $\tilde{r}_j < \tilde{r}_0$ where $\tilde{r}_0$ defines the instability boundary such that $\psi^*(\tilde{r}_0) = 0$. This boundary radius is always greater than r_1 , and r_0 may be computed by evaluating the index m where

$$\psi^*(\tilde{r}_{m-1}) \geq 0, \quad \psi^*(\tilde{r}_m) < 0. \quad (18)$$

We similarly perform linear interpolation to obtain $\tilde{r}_0$,

$$\tilde{r}_0 = \tilde{r}_{m-1} - \frac{\psi^*(\tilde{r}_{m-1})}{\psi^*(\tilde{r}_m) - \psi^*(\tilde{r}_{m-1})}. \quad (19)$$

As $\tilde{r}_1$ and $\tilde{r}_0$ are obtained, $\tilde{r}_-$ and $\tilde{r}_+$ are evaluated such that

$$\tilde{r}_- < \tilde{r}_0, \quad (20)$$

$$\tilde{r}_0 < \tilde{r}_+ < \tilde{r}_0, \quad (21)$$

and the reconnection of field lines is considered, which follows

$$\psi^*(\tilde{r}_-) = \psi^*(\tilde{r}_+). \quad (22)$$

The post-crash profiles where $\tilde{r}_i < \tilde{r}_0$ are determined by

$$q_f(\tilde{r}_j) = 1, \quad (23)$$

$$B_{\theta,f}(\tilde{r}_j) = \tilde{r} \epsilon_a B_0, \quad (24)$$

$$p_f(\tilde{r}_i) = p_i(\tilde{r}_0) = p_0(1 - \tilde{r}_0^2)^\mu. \quad (25)$$